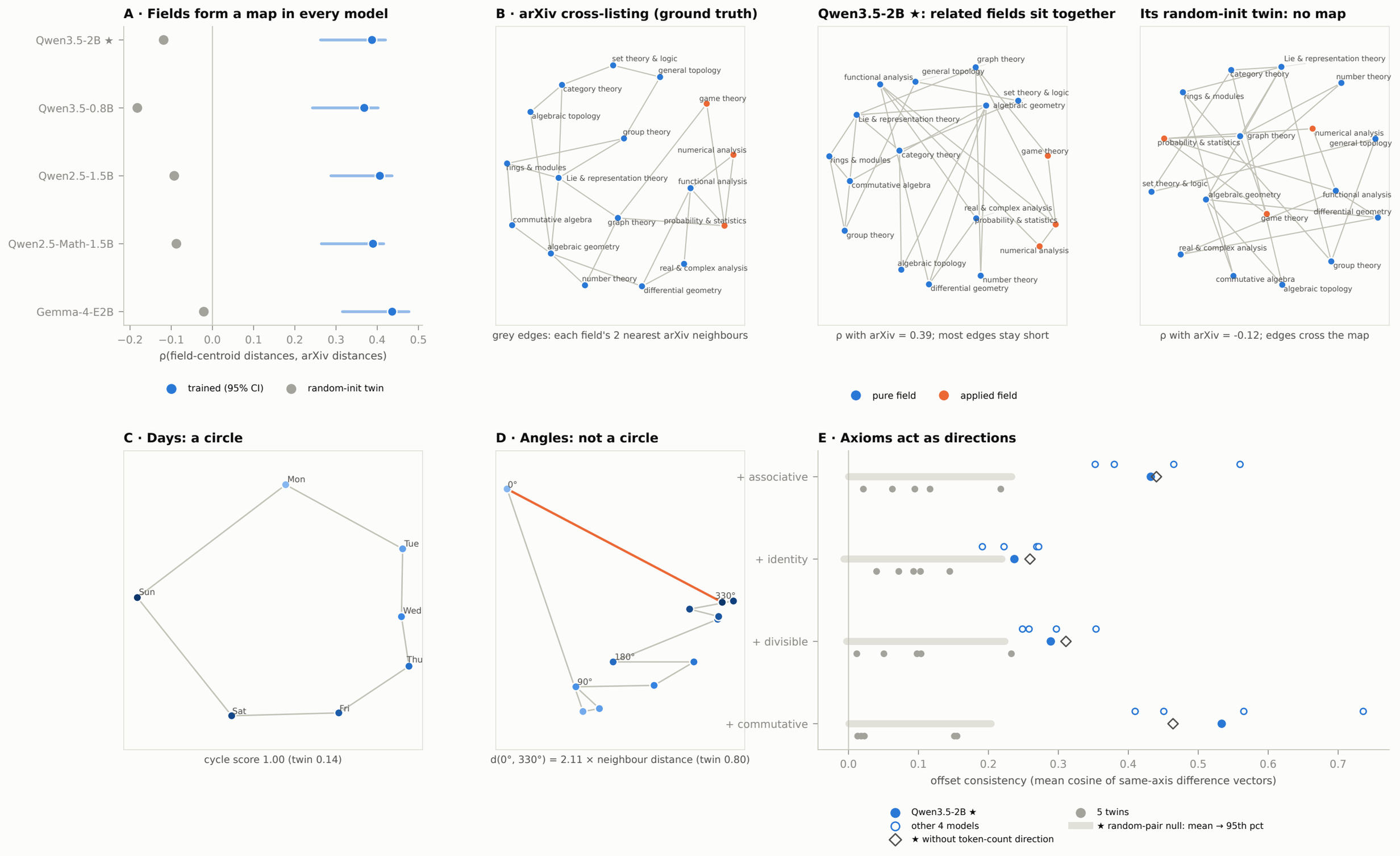
A · Fields form a map in every model
Qwen3.5-2B ★
Qwen3.5-0.8B
Qwen2.5-1.5B
Qwen2.5-Math-1.5B
Gemma-4-E2B
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
ρ(field-centroid distances, arXiv distances)
trained (95% CI)
random-init twin
B · arXiv cross-listing (ground truth)
set theory & logic
general topology
category theory
game theory
algebraic topology
group theory
numerical analysis
rings & modules
Lie & representation theory
functional analysis
commutative algebra
graph theory
probability & statistics
algebraic geometry
real & complex analysis
number theory
differential geometry
grey edges: each field's 2 nearest arXiv neighbours
Qwen3.5-2B ★: related fields sit together
graph theory
general topology
functional analysis
set theory & logic
algebraic geometry
Lie & representation theory
rings & modules
category theory
game theory
commutative algebra
real & complex analysis
probability & statistics
group theory
numerical analysis
algebraic topology
number theory
differential geometry
ρ with arXiv = 0.39; most edges stay short
Its random-init twin: no map
Lie & representation theory
category theory
number theory
rings & modules
numerical analysis
graph theory
probability & statistics
general topology
set theory & logic
algebraic geometry
functional analysis
game theory
differential geometry
real & complex analysis
group theory
commutative algebra
algebraic topology
ρ with arXiv = -0.12; edges cross the map
pure field
applied field
C · Days: a circle
Mon
Tue
Wed
Thu
Fri
Sat
Sun
cycle score 1.00 (twin 0.14)
D · Angles: not a circle
0°
90°
180°
330°
d(0°, 330°) = 2.11 × neighbour distance (twin 0.80)
E · Axioms act as directions
+ associative
+ identity
+ divisible
+ commutative
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
offset consistency (mean cosine of same-axis difference vectors)
Qwen3.5-2B ★
other 4 models
★ without token-count direction
5 twins
★ random-pair null: mean → 95th pct